TECHNICAL PROPOSAL

FY2027 Department of War Defense University Research Instrumentation Program (DURIP)

Office of Naval Research · Notice of Funding Opportunity N0001426SF003

High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility for Time-Resolved Thermometry and Energy-Transfer Kinetics in Naval Propulsion Flows

Item	Entry
Applicant institution	Morgan State University, Clarence M. Mitchell, Jr. School of Engineering, 1700 East Cold Spring Lane, Baltimore, MD 21251. Historically Black University; Minority-Serving Institution. UEI KULSKCCZJT27; CAGE 8U547
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Agency and program officer (SF-424 R&R fields 4a and 4b)	N00014; 352 [Marineau, Eric]. Code 352, Hypersonic Aerothermodynamics, High-Speed Propulsion and Materials
Amount requested	**$1,067,994.** Equipment $1,037,471 (Module A, fs-CARS diagnostic, $915,606; Module B, reference-flame facility, $121,865), exempt from indirect cost; acceptance-test travel $3,500; extended warranty $16,320; indirect cost $10,703. No cost share.
Basis of cost	Written vendor quotations for items 1, 2, 6, 7, 8, 9 and 18 and for the extended warranty ($536,701, half of the request), partial quotations for items 10, 11, 14, 15 and 16, and documented budgetary estimates for the remaining items (Budget Justification, Table B-2)
Period of performance	12 months, 18 June 2027 to 17 June 2028
Date of submission	November 2026

Project Abstract

APPROVED FOR PUBLIC RELEASE

Title: High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility for Time-Resolved Thermometry and Energy-Transfer Kinetics in Naval Propulsion Flows

Principal Investigator: Dr. Oluwatobi Busari, Morgan State University. **Program Officer:** Dr. Eric Marineau, Office of Naval Research, Code 352.

Morgan State University requests $1,067,994 for one instrumentation system in two modules. Module A is a high-repetition-rate femtosecond coherent anti-Stokes Raman scattering (fs-CARS) diagnostic: a 40 W ytterbium femtosecond amplifier operating at 100 kHz, a tunable optical parametric amplifier (650 to 900 nm), a dual-channel programmable pulse shaper, an imaging spectrograph, and a high-speed camera with a gated image intensifier. Module B is the sub-atmospheric reference-flame facility the diagnostic measures: a Hencken burner, a 300 mm optically accessible vessel held between 760 and 20 Torr, hydrogen capability, and a rotating-detonation injector element.

Femtosecond/picosecond CARS instruments in use today run at 1 kHz. They resolve temperature fluctuations only up to 500 Hz and sample a 0.58 ms nitrogen vibrational relaxation (2000 K, 1 atm) less than once. Burst-mode lasers reach higher rates, but only for a few milliseconds at a time. The requested system records one complete spectrum every 10 microseconds, a hundred times faster, so temperature becomes a continuous time series that resolves fluctuations up to 50 kHz. A gas parcel crossing the probe volume at 3 m/s is measured about four times, and a 5 to 20 kHz rotating-detonation cycle is sampled 5 to 20 times. Programmable pulse shaping adds an adjustable chirped probe (0.5 to 5 ps), selection of the molecules that are excited, and pulse trains timed to rotational revivals. First-principles estimates, checked against the PI's density-matrix CARS model, give single-shot spectra with a signal-to-noise ratio of 30 or better from 760 Torr down to about 320 Torr, and 45 to 90 averaged temperatures per second at 20 Torr. The tunable pump reaches N_2, O_2, H_2, CO, CH_4 and H_2O.

The instrument supports four lines of research: high-rate thermometry of unsteady, lifted and low-pressure flames, ignition transients and rotating-detonation injectors; energy-transfer and reaction kinetics, including vibrational non-equilibrium at low pressure and hydrogen kinetics, as experimental tests of computed potential energy surfaces; shaped-pulse CARS methods for selective, low-background detection; and validation of passive optical thermometry, with high-speed imaging of flames and rocket static fires. This work serves Navy needs in non-intrusive, time-resolved diagnostics for high-speed propulsion, thermochemical non-equilibrium in hypersonic flow, pressure-gain combustion, and infrared plume signatures at altitude. The research is the subject of a pending ONR proposal by the PI, which requests no equipment and relies on Module B.

Morgan State University, Maryland's Preeminent Public Urban Research University, will install the system in a dedicated laser laboratory and open it to other investigators. Students who install and operate it will gain hands-on experience with ultrafast lasers, nonlinear spectroscopy and high-rate data analysis, skills used in DoW propulsion and diagnostics laboratories.

1. Technical Approach

1.1 Summary of the request

Morgan State University (MSU) requests **$1,067,994** to acquire, install and commission one instrumentation system built in two modules (Table 2). **Module A** is a high-repetition-rate femtosecond coherent anti-Stokes Raman scattering (fs-CARS) diagnostic with programmable pulse shaping. Its core is a 40 W ytterbium (Yb) femtosecond amplifier running at 100 kHz (selectable to 200 kHz and above), a tunable optical parametric amplifier (OPA), a dual-channel 4f pulse shaper, a 320 mm imaging spectrograph, and a high-speed CMOS camera with a gated image intensifier, together with the timing, delay, beam-delivery, enclosure and data systems that make them one instrument ($915,606). **Module B** is the sub-atmospheric reference-flame facility that the diagnostic measures: an atmospheric Hencken burner and gas panel, a 300 mm optically accessible vessel with closed-loop pressure control from 760 to 20 Torr, hydrogen capability, a single-element rotating-detonation injector, a spectrometer-based wavelength reference and volume-Bragg-grating collection filters ($121,865). Equipment is $1,037,471, or 97 percent of the request, and is exempt from indirect cost. The balance covers one acceptance-test trip and a second-year extended warranty on the laser and OPA, with their indirect cost. Every component serves a single research purpose, the time-resolved, non-intrusive measurement of temperature and molecular state populations in reacting and non-equilibrium flows from sea-level pressure to the altitude regime. The two modules therefore form one system in the sense of NOFO Section b.3.i.

The core of the system is priced from written vendor quotations received in September and October 2026: the laser and OPA (Light Conversion), the gated intensifier (Tibidabo Scientific, U.S. distributor for Lambert Instruments), the delay stage (Zaber), the delay generator (Berkeley Nucleonics), the optical tables (Newport) and the Bragg-grating filters (OptiGrate), together with the extended warranty. These quotations total $536,701, half of the request. Partial quotations cover parts of five further items. The remaining items carry budgetary estimates, and their quotations are listed with their status in Table B-2 of the Budget Justification.

The instrument gives MSU three capabilities it does not have today. (1) **A measurement rate one hundred times higher** than the 1 kHz femtosecond/picosecond CARS systems that define the current state of the art [1–3]: one complete CARS spectrum every 10 µs (5 µs at 200 kHz), so that temperature becomes a continuous time series that resolves fluctuations up to 50 kHz. (2) **Programmable control of the pulses on every shot**: an adjustable chirped probe (0.5 to 5 ps), selection of the molecules that are excited, and shaped pulse trains [4]. (3) **Single-shot spectra from 760 Torr down to about 320 Torr**, and averaged temperatures at 45 to 90 values per second at 20 Torr, using a gated intensified camera that also rejects flame luminosity [5]. Module B supplies what the diagnostic needs at altitude-representative conditions: flames of independently known temperature from 760 to 20 Torr, hydrogen–air flames, and a rotating-detonation injector element at matched injection stiffness. Module B is also the test-article hardware of the PI's pending ONR proposal, which requests no equipment and names this DURIP request as the source of that hardware (Section 1.4).

The narrative follows the three equally weighted DURIP evaluation criteria: impact on research that DoW funds or plans to fund (Sections 1.2 to 1.5 and 1.13), importance to DoW missions (Section 1.6), and the education of future scientists and engineers (Section 1.12). Sections 1.9 to

1.11 address the interface with existing facilities, special acquisition and installation circumstances, and useful life, as NOFO Section d.1.iv.C requires.

1.2 The measurement gap

Temperature is the state variable that governs reaction rates, pollutant formation, radiative signatures and thermal loads in propulsion systems. CARS is the reference technique for measuring it in flames because its signal is a coherent, laser-like beam whose spectrum records the population of every rotational and vibrational level of the probed molecule [1, 6]. Hybrid fs/ps CARS made single-shot thermometry possible at 1 kHz [2] and is now used for kHz-rate measurements in canonical and practical flames [3]. A thousand spectra per second is still slow compared with the processes that matter in naval propulsion flows. Table 1 compares their time scales with the sampling that a 1 kHz system and the requested 100 kHz system provide.

Table 1. *Characteristic times of target processes and the number of CARS spectra recorded per characteristic time.*

Process	Characteristic time	Spectra at 1 kHz	Spectra at 100 kHz
Gas parcel crossing the 110 µm probe volume at the 3.0 m/s exit velocity of the PI's burner	37 µs	0.04	3.7
One cycle of a rotating-detonation wave at 5–20 kHz [10]	50–200 µs	0.05–0.2	5–20
N_2–N_2 vibrational relaxation, 2000 K, 1 atm (Millikan–White [7])	0.58 ms	0.6	58
Test time of a conventional reflected-shock tube (typical)	1–2 ms	1–2	100–200

At 1 kHz each of these processes is sampled less than about once per characteristic time. Its dynamics must then be inferred from statistics over many independent events, and fluctuations above 500 Hz are not resolved at all. At 100 kHz each process is sampled 4 to 200 times, and its dynamics are measured directly within a single event. Burst-mode lasers reach 100 kHz and higher, but only in bursts of a few milliseconds separated by seconds, and with fixed pulse formats. Nanosecond CARS at 10 Hz is slower still and is limited by non-resonant background and collisional effects [1]. No continuous 100 kHz CARS instrument with programmable pulses exists at MSU, and this request would provide one.

1.3 Instrument description and predicted performance

Module A architecture. The Yb amplifier (item 1 in Table 2; Light Conversion CARBIDE-40W) delivers 400 µJ pulses of less than 250 fs at 1030 nm and 100 kHz (40 W), with the repetition rate adjustable from 100 kHz to 1 MHz through an integral pulse picker. Most of its output pumps the OPA (item 2; ORPHEUS-F), whose signal is tunable from 650 to 900 nm with durations below 55 fs at 800–900 nm and below 70 fs at 650–800 nm. The OPA signal provides the broadband CARS pump and probe. A portion of the narrower-band 1030 nm fundamental is split off before the OPA and serves as the Stokes beam; the vendor recommends this arrangement because the 1030 nm light leaving the OPA is depleted. The pump and probe pass through the two channels of a reflective 4f pulse shaper built around a 1272 × 1024 liquid-crystal-on-silicon (LCoS) spatial light modulator (item 3). On the pump channel an amplitude mask selects the excitation spectrum; on the probe channel a programmed spectral phase sets a chirped probe of 0.5 to 5 ps. The shaper window is 18.3 ps with 367 independent

spectral features [5]. A 300 mm motorized stage (item 7) sets the probe delay, and an 8-channel digital delay generator (item 8) synchronizes the laser, the intensifier gate and the camera. The CARS beam is dispersed by a 320 mm aberration-corrected imaging spectrograph with a 1200 lines/mm grating (item 4) onto a high-speed CMOS camera with 512 GB of memory (item 5). The camera is read out over a narrow spectral region of interest, so that one spectrum is recorded on every laser shot at 100 kHz, or at 200 kHz when fluctuations up to 100 kHz must be resolved; at that region of interest the memory holds continuous records of several tens of seconds. A gated GaAs image intensifier (item 6; Lambert HiCATT25 with a 25 mm photocathode, 3 ns minimum gate, 300 kHz gating, 1:1 relay optics and F-mount input and output) adds gain at low pressure and gates out flame luminosity. Two pneumatically isolated optical tables (item 9) carry the source and the CARS and detection optics. Beam-delivery optics, a power meter and a beam profiler (item 10), and a HEPA-filtered, laser-safe enclosure with laser-safety curtains (item 11) complete the optical system. A GPU workstation (item 12) streams camera data and fits spectra.

Module B, the reference-flame facility. Module B comprises the following components. Item 13 is an atmospheric 50 mm Hencken burner with gas panel, pressure instrumentation and wavelength reference, which provides the atmospheric anchor for every measurement. Item 14 is a 300 mm inner-diameter, 600 mm stainless full-vacuum vessel with seven port flanges: four orthogonal 63.5 mm clear-aperture UV-fused-silica windows, two ports at ±20° for crossed-beam diagnostics such as CARS, and an axial port. It is served by a 100 m^3/h chemical-duty dry screw pump, a throttle valve under closed-loop control from 760 to 20 Torr, capacitance manometers, an exhaust heat exchanger, a revised reduced-pressure burner with choked micro-tubes and a quartz liner, and a safety layer of UV flame detection, interlock and nitrogen purge. Item 15 adds hydrogen capability: mass-flow control, a two-stage regulator and excess-flow valve, two-sensor leak detection, a gas cabinet and a purge panel. Item 16 is a modular single-element rotating-detonation injector head in three geometries, with high-response plenum transducers and a sapphire injector-face window. Item 17 is a spectrometer-based wavelength reference for the OPA and for atomic emission lines, and item 18 is a set of volume-Bragg-grating notch filters at 620, 700 and 780 nm for the collection optics designed by Dr. Albert O. Okorogu. At the design point (50 Torr, 3.0 m/s, 75.4 SLPM, 1.32 kW) the flame is thick enough to be probed with millimetre resolution yet one-dimensional over the burner core. Module B is a fabricated system assembled at MSU, and its components are budgeted as equipment.

Table 2. *System components in two modules, one system for a common research purpose (NOFO b.3.i). Basis: Q = written vendor quotation (number and date in Table B-2); P = partial quotation; E = budgetary estimate with a quotation requested.*

#	Component (specification)	Role in the measurement	Cost	Basis
	Module A: 100 kHz fs-CARS diagnostic			
1	Yb femtosecond amplifier, Light Conversion CARBIDE-40W-0.4mJ: 1030 nm, 40 W, 400 µJ at 100 kHz, < 250 fs, 100 kHz–1 MHz, pulse picker, chiller; delivery, on-site installation and training; estimated import duty	Drives the OPA; provides the Stokes beam	$220,904	Q
2	Yb-pumped OPA, Light Conversion ORPHEUS-F: 650–900 nm signal, 1200–2500 nm idler, < 55 fs at 800–900 nm; estimated import duty	Tunable broadband pump and probe	$76,164	Q
3	Dual-channel reflective 4f pulse shaper: LCoS	Pump spectral mask;	$35,000	E

	modulator 1272 × 1024 (Hamamatsu X15213-02 class, 800 ± 50 nm), gratings, optics, mounts	probe chirp 0.5–5 ps; pulse trains		
4	Imaging spectrograph, 320 mm, aberration-corrected, 1200 l/mm grating, F-mount	Disperses the CARS signal	$40,000	E
5	High-speed CMOS camera, 512 GB (Phantom TMX 7510 class)	One spectrum per shot at 100–200 kHz	$264,000	E
6	Gated GaAs image intensifier, Lambert HiCATT25 (25 mm, 3 ns gate, 300 kHz, 1:1 relay, F-mount) with IC-GC-USB gate control; estimated tariff	Gain at low pressure; luminosity gating	$152,531	Q
7	Motorized delay stage, Zaber X-LDQ0300C, 300 mm, built-in controller	Probe delay	$8,689	Q
8	Digital delay generator, 8 channels, Berkeley Nucleonics 577-8C	Timing of laser, gate, camera	$7,583	Q
9	Optical tables, Newport RS2000 5 × 10 ft and 4 × 8 ft, two S-2000A pneumatic isolator sets	Source table; CARS and detection table	$37,735	Q
10	Beam-delivery optics and optomechanics, 50 W power meter, beam profiler, laser-safety eyewear	Beam routing, overlap and diagnostics	$45,000	P
11	Laser-table enclosure with HEPA fan-filter units and laser-safe panels; free-standing laser-safety curtains	Optics protection; beam containment	$16,000	P
12	GPU acquisition and spectral-fitting workstation	Data streaming and fitting	$12,000	E
	Module A subtotal		**$915,606**	
	Module B: sub-atmospheric reference-flame facility			
13	Atmospheric Hencken burner (50 mm), gas panel with mass-flow controllers, pressure instrumentation, wavelength reference	Atmospheric anchor flame (1600–2400 K)	$4,690	E
14	Reduced-pressure vessel (300 mm ID × 600 mm, 304 SS, 7 ports, 6 UVFS windows), dry screw pump, throttle valve and controller, capacitance manometers, exhaust heat exchanger, vacuum plumbing, revised burner, flame detector, interlock and N_2 purge	Reference flames at 760–20 Torr with optical access for CARS	$61,300	P
15	Hydrogen capability: H_2 mass-flow controller, two-stage regulator, excess-flow valve, two-sensor leak detection, gas cabinet, purge panel	H_2–air reference flames	$11,500	P
16	Single-element RDC injector head (17-4 SS, 3 geometries), high-response plenum transducers, sapphire injector-face window	Injector response at matched stiffness	$13,600	P
17	Spectrometer-based wavelength reference: high-resolution fiber spectrometer (500–1100 nm) with Ne/Ar/Hg calibration lamps	Wavelength reference for OPA tuning and emission lines	$14,000	E
18	Volume-Bragg-grating notch filters, OptiGrate BragGrate 620, 700 and 780 nm, OD4, AR coated (Okorogu collection-optics design)	Narrow-band rejection; high-étendue collection	$16,775	Q
	Module B subtotal		**$121,865**	
	Equipment total		**$1,037,471**	

Predicted performance. First-principles estimates for the products of a stoichiometric methane–air flame [5] give about 8.5×10^4 CARS photons per shot at 760 Torr. With the camera

alone, single-shot N_2 spectra reach a signal-to-noise ratio (SNR) of about 36 at 760 Torr. With the intensifier, single-shot spectra with SNR of 30 or better extend down to about 320 Torr, and below that pressure averaging gives 45 to 90 temperatures per second at 20 Torr. The coherent Raman signal scales with the square of number density, so at 20 Torr it is 6.9×10^{-4} of its atmospheric value; the 100 kHz rate is what keeps averaged measurements practical there. An independent check with the PI's density-matrix CARS model reproduced the excitation and coherence calculation to within 5 percent. The remaining uncertainty, about a factor of three, lies in the Raman cross section and the usable laser intensity, so the signal level is the first quantity measured at commissioning (benchmark G4, Section 1.8).

Species coverage. With the Stokes beam at 1030 nm, tuning the OPA pump from 721 to 888 nm reaches the Q-branches of H_2, H_2O, CH_4, N_2, CO and O_2 (Table 3). The CO_2 Fermi dyad lies just outside the OPA range. N_2 and CO can be excited together by shaping the pump spectrum (Section 1.5, Thrust 3).

Table 3. *Species reachable with the selected OPA (Stokes 1030 nm).*

Molecule	Raman shift (cm^{-1})	Pump (nm)	Signal (nm)	Reachable
H_2 Q(1)	4155.25	721.3	555.0	Yes
H_2O	3657	748.2	587.4	Yes
CH_4	2917	792.0	643.4	Yes
N_2 Q	2329.9	830.7	696.0	Yes
CO Q	2143	843.7	714.5	Yes
O_2 Q	1556	887.7	779.9	Yes (filter change)
CO_2 Fermi dyad	1388 / 1285	901.1 / 909.6	801.0 / 814.4	No (just outside)

1.4 Innovation relative to alternative approaches

Yb amplifiers, OPAs, LCoS shapers and high-speed cameras are each commercially mature. The innovation is their combination into a continuous 100 kHz CARS instrument whose pump and probe can be reprogrammed in software from one experiment to the next. Because each subsystem is mature, the technical risk lies in integration, and the acceptance benchmarks of Section 1.8 are written to retire that risk. Table 4 compares the system with the alternatives.

Table 4. *The requested system compared with existing CARS approaches.*

Approach	Repetition rate	Duty cycle	Pulse control	Limitation for this research
Nanosecond broadband CARS	10 Hz	Continuous	Fixed	Shots 100 ms apart; non-resonant background
Hybrid fs/ps CARS, Ti:sapphire [2, 3]	1 kHz	Continuous	Fixed probe	Samples the Table 1 processes less than once
Burst-mode ps CARS	100 kHz and above, in bursts	Millisecond bursts, seconds apart	Fixed	Short records; no shaping
This request: Yb fs-CARS with 4f shaper	**100 kHz (200 kHz)**	**Continuous**	**Pump mask, probe chirp, pulse trains**	**None of the above**

Relation to the PI's ONR proposal. The PI's pending ONR Long Range BAA proposal,

Altitude Compensated Combustion Diagnostics (ACCD) Laboratory ($658,586 over 36 months, Code 352; Section 1.13), funds the people and consumables of the research: PI and Co-PI summer effort, two graduate research assistants, materials, travel and publication. It requests no equipment and names this DURIP request as the source of its test-article hardware, which is Module B. The two requests are complementary. ONR would pay for the research effort and DURIP for the instrument and the facility, and no item or cost appears in both. If this DURIP request is not funded, the ONR plan limits its first year to atmospheric work and seeks the Module B hardware through an award modification.

1.5 Research the instrument will enable

Thrust 1: High-repetition-rate thermometry in unsteady and low-pressure flames

At 100 kHz the gas moves 30 μm between shots at the 3.0 m/s exit velocity of the PI's burner, about one quarter of the 110 μm probe-volume width, so each fluid parcel is measured about four times as it passes. Temperature becomes a continuous time series that resolves fluctuations up to 50 kHz (100 kHz at 200 kHz), compared with 500 Hz for a 1 kHz system. Planned studies are: (a) the 27-condition reference ladder of the ONR program (1600 to 2400 K at six pressures from 760 to 20 Torr) in the Module B vessel, with liftoff, flicker and blow-off of its flames recorded continuously while the vessel is pumped down; (b) the growth of a laser-ignition kernel into a flame, shot by shot; (c) rotating-detonation injector studies on the Module B single-element injector at matched injection stiffness (100 Torr plenum against a 50 Torr chamber), and in detonation cycles of 5 to 20 kHz, where 100 kHz gives 5 to 20 samples per cycle and 200 kHz gives 10 to 40, enough to phase-resolve injector recovery; and (d) static versus total temperature in compressible reacting flows, as a direct test of the PI's published finding that the combustor exit Mach number, through the static temperature, governs NO_x formation [8].

Thrust 2: Energy-transfer and reaction kinetics

Rate coefficients for vibrational energy transfer and for elementary reactions are now computed routinely on ab initio potential energy surfaces by quasi-classical trajectory and master-equation methods. The experimental test of those surfaces is the distribution of molecules over vibrational and rotational levels and its evolution in time, which is what one broadband fs-CARS spectrum records for a Raman-active molecule. (a) **Vibrational relaxation.** The Millikan–White correlation [7] gives 0.58 ms for N_2–N_2 relaxation at 2000 K and 1 atm. A 1 kHz system samples it 0.6 times; the requested system samples it 58 times. Collisions with water shorten relaxation in combustion products, which makes the higher rate more necessary. (b) **Vibrational non-equilibrium at low pressure.** The same correlation gives 22 ms at 2000 K and 20 Torr, longer than the residence time of a few milliseconds of gas leaving the low-pressure flames of Module B. N_2 vibrational and rotational temperatures may therefore differ measurably, and a chirped-probe spectrum separates the hot band from the fundamental and reads both temperatures in one shot. (c) **Hydrogen kinetics.** In the hydrogen–air flames of Module B, H_2 Q-branch spectra resolve the v = 0, 1 and 2 populations and the rotational distribution, which are the observables against which H_2 energy-transfer and H_2–O_2 reaction models are compared. (d) **Impulsively initiated chemistry.** With a pulsed initiator such as shock heating, laser photolysis or a nanosecond discharge (none of which is part of this request), each event yields a time history sampled every 10 μs: 100 to 200 spectra in 1 to 2 ms of shock-tube test time instead of one or two. For events that cannot be repeated cheaply, the higher rate decides whether the

measurement is possible at all.

Thrust 3: Shaped-pulse CARS methods

(a) **Adjustable chirped probe.** In a 2226 K flame a 1.5 ps probe uses 11.4 percent of its energy, against 6.1 percent for a 3 ps probe typical of established hybrid fs/ps systems [2, 3], which nearly doubles the low-pressure signal [5]. (b) **Selective excitation.** Because the Stokes pulse is narrower in bandwidth than the pump, a mask on the pump spectrum acts directly as a mask on the excitation spectrum, with a resolution of about 59 cm^{-1}. N_2 thermometry and CO detection can be combined in rich flames, and spectral components that add only non-resonant background can be removed. (c) **Pulse trains.** Two-pulse trains spaced at the N_2 rotational revival time (8.38 ps, against 11.60 ps for O_2) favor N_2 over O_2 rotational coherence, an exploratory route to local mixture composition. (d) **Coherent preparation.** The design pump and Stokes pulses transfer about 0.17 percent of the N_2 molecules in the focal volume to v = 1 per shot, a small but measurable population for pump–probe studies of relaxation. Pulse shaping adds selectivity and control; the measurement rate is set by the repetition rate alone.

Thrust 4: Validation of passive thermometry and high-speed imaging

(a) **Reference for passive optical thermometry.** The PI's passive coherence-thermometry concept, proposed separately (Section 1.13), requires validation against N_2 CARS, and this system provides that reference in the same laboratory. (b) **High-speed imaging.** Used without the laser, the camera and intensifier record flame chemiluminescence, flame liftoff and injector faces, and liquid-rocket static fires in the PI's rocketry program funded by the State of Maryland. (c) **Laser ignition (exploratory).** Focused to 10 μm, the 400 μJ, 250 fs pulses reach about 10^{15} W cm^{-2}, above typical air-breakdown intensities, so the amplifier may also serve as an ignition source in the Module B vessel; this will be tested during commissioning. (d) **Plume-signature validation data.** CARS temperatures measured in the Module B flames from 760 to 20 Torr are the reference against which the ONR program compares infrared plume-signature radiative-transfer models in the Doppler-limited regime of high-altitude flight.

1.6 Future naval relevance and importance to DoW missions

ONR Code 352 (Hypersonic Aerothermodynamics, High-Speed Propulsion and Materials), to which this proposal is addressed, lists among its interests "novel flow diagnostics for non-intrusive, time-resolved measurements of velocity and thermodynamic state-variables" and experimental studies with advanced diagnostics of solid-fuel ramjet combustion [9]. A 100 kHz CARS thermometer is that diagnostic for temperature. The instrument serves naval missions in five areas. (1) **High-speed and air-breathing propulsion:** time-resolved temperature and species in unsteady, lifted and igniting flames, the regimes that set combustor operability. (2) **Thermochemical non-equilibrium:** direct measurement of vibrational and rotational temperatures and of vibration–translation relaxation, processes that hypersonic aerothermodynamic models must represent. (3) **Pressure-gain combustion:** phase-resolved thermometry of rotating-detonation injector recovery [10], also of interest to Code 351 (Power, Propulsion and Thermal Management). (4) **Plume signatures at altitude:** temperature measured, rather than inferred, in the Module B reference flames from 760 to 20 Torr, the static pressures of flight from sea level to about 25 km [11], to anchor the infrared signature models used for detection, tracking and discrimination. (5) **Hydrogen and alternative fuels:** benchmark

H_2 thermometry and kinetics for fuel-flexible naval power.

The same research is relevant to the Army Research Office Propulsion and Energetics program and to the Air Force Office of Scientific Research portfolio in combustion and non-equilibrium thermodynamics. All work is fundamental research performed on campus. Results will be published without restriction, and no ITAR- or EAR-controlled technical data are anticipated.

1.7 Project schedule and milestones

The award period is 12 months from the anticipated start date of 18 June 2027, and purchase orders for every item in both modules are placed in Month 1. The quoted lead times set the sequence: 6 to 8 months after order for the laser and OPA (Light Conversion), 22 to 26 weeks for the intensifier (Tibidabo Scientific), 24 weeks for the injector pressure transducers (Kulite), 10 weeks for the vessel windows (Crystran), 4 to 8 weeks for the optical tables and enclosure (Newport), 4 to 6 weeks for the power meter and beam profiler (Gentec-EO) and a few days for the delay stage (Zaber). The laser and OPA therefore arrive in Months 7 and 8 and are installed, aligned and accepted on site by the vendor, whose quotation includes installation and operator training. The detection subsystem and the pulse-shaper optics are assembled and tested while the laser is being built, so that first CARS spectra follow within a month of laser acceptance. In Module B the vessel and pump are the long-lead items and are installed by Month 10, and hydrogen is not introduced until micro-tube choking has been verified. Table 5 shows the work by month and Table 6 the milestones. If delivery slips beyond the float in this plan, MSU will request the 12-month no-cost extension that the NOFO allows; the installation sequence would not change.

Table 5. *Installation and commissioning schedule (months from 18 June 2027; shaded cells mark active months).*

Task	1	2	3	4	5	6	7	8	9	10	11	12
A. Orders placed; Lab 229 prepared by MSU	■	■	■									
B. Tables, enclosure, timing, delay stage, workstation installed		■	■	■								
C. Spectrograph and camera bench-tested; intensifier received (22–26 weeks)			■	■	■	■	■					
D. Laser and OPA delivered (6–8 months), installed and accepted by the vendor							■	■	■			
E. Pulse shaper assembled; calibrated with the OPA						■	■	■	■			
F. First CARS spectra; 100 kHz benchmarks									■	■	■	
G. First unsteady and low-pressure CARS datasets; final report										■	■	■
H. Module B: Hencken burner, gas panel, references commissioned	■	■	■									
I. Module B: vessel, pump, pressure control built; choking verified	■	■	■	■	■	■	■	■	■	■		
J. Module B: hydrogen system and injector element (transducers 24 weeks) installed							■	■	■	■	■	■

Table 6. *Milestones.*

No.	Milestone	Month	Date
M1	Purchase orders placed for all items in both modules	1	Jul 2027

M2	Lab 229 ready (circuits, interlock, tables, enclosure); Hencken burner and gas panel commissioned	3	Sep 2027
M3	Detection subsystem (spectrograph, camera, intensifier, timing) accepted	7	Jan 2028
M4	Laser and OPA accepted on site against specification (G1)	9	Mar 2028
M5	Pulse shaper calibrated (G2)	9	Mar 2028
M6	First N_2 CARS spectra recorded on every shot at 100 kHz (G3)	10	Apr 2028
M7	Vessel installed; micro-tube choking verified before hydrogen	10	Apr 2028
M8	Signal and temperature-accuracy benchmark in the atmospheric Hencken flame (G4)	11	May 2028
M9	Vessel in closed-loop operation from 760 to 20 Torr; first low-pressure CARS dataset; hydrogen system and injector installed; shared use opened; final report (G5, G6)	12	Jun 2028

1.8 Goals and benchmarks

The goal of the award is a commissioned system, both modules, that meets its specification and is in research use by the end of the 12-month period. Progress is measured against the six benchmarks of Table 7. Each is checked at the milestone shown and reported to the Program Officer.

Table 7. *Goals and benchmarks of success.*

No.	Goal	Benchmark of success	Month
G1	Source performance	Laser 40 W and 400 μJ at 100 kHz, < 250 fs; OPA tunable 650–900 nm, < 55 fs at 800–900 nm; verified in the vendor's site acceptance test	9
G2	Pulse control	Probe chirp set from 0.5 to 5 ps and pump spectral mask at about 59 cm^{-1} resolution, verified by cross-correlation and spectral measurement	9
G3	Measurement rate	N_2 CARS spectrum recorded on every shot at 100 kHz in a continuous record of at least 1 s	10
G4	Signal and accuracy	Single-shot SNR ≥ 30 at 760 Torr in methane–air products; mean temperature in the atmospheric Hencken flame within ±41 K of adiabatic equilibrium, the reference-flame uncertainty at 760 Torr [12]	11
G5	Low pressure	Vessel holds set-points from 760 to 20 Torr in closed loop; first averaged CARS temperatures at or below 100 Torr	12
G6	Use and training	Laser-safety SOP approved; at least 2 graduate and 4 undergraduate students trained; shared-use calendar open	12

1.9 Interface with existing facilities and instruments

Location. Module A will be installed in Room 229 of the Clarence M. Mitchell, Jr. School of Engineering, the PI's laboratory. The room is windowless and already fitted with laser barrier curtains, and it is preferred over Room 136, where machining raises dust and vibration. Room 229 accommodates both optical tables of this system (5 × 10 ft and 4 × 8 ft) and the passive-thermometry instruments proposed separately (Section 1.13).

Existing instruments. The system connects to the PI's existing laboratory and rocketry-program resources: the data-acquisition hardware, pressure and temperature instrumentation, and gas-handling equipment of the Morgan State rocketry program, and the School of Engineering

machine shop that fabricates mounts, the Hencken burner and the injector head. The atmospheric Hencken burner of Module B (item 13), on the Room 229 optics bench, provides the commissioning flame for benchmarks G3 and G4.

Module B vessel. Its four orthogonal 63.5 mm ports and two ports at ±20° were designed for crossed-beam diagnostics such as CARS, so the Module A beams enter it without modification. The vessel, pump and gas panel are built on a mobile frame. CARS campaigns run with the vessel beside the CARS table in Room 229, with the pump exhaust ducted to the building exhaust and hydrogen supplied from the ventilated gas cabinet. Injector testing and long-duration hydrogen operation take place at the university's Lake Clifton site, which provides exhaust, pump-noise isolation and hydrogen ventilation.

Other links. The camera and intensifier are portable and will also be used at Lake Clifton for rocket static-fire imaging. Data stream to the dedicated workstation (item 12) and are archived on university-managed, backed-up storage.

1.10 Special acquisition and installation circumstances

Long lead times. The laser and OPA are built to order, with a quoted delivery of 6 to 8 months after order; the intensifier is quoted at 22 to 26 weeks and the injector transducers at 24 weeks. All are ordered at award (Section 1.7).

Vendor installation and delivery. The Light Conversion quotation includes delivery to the loading dock, on-site installation and alignment, and operator training, with the engineer's travel and expenses. The Newport optical tables (3,980 lb shipping weight) are delivered to the dock only; MSU Facilities will move them into Room 229 using a local rigging contractor, at university expense. The intensifier is quoted Ex-Works Madison, Connecticut, and freight will be confirmed with the vendor at the time of order.

Tariffs and import duty. The laser and OPA are manufactured in Lithuania. The Light Conversion quotation states that an import duty, currently 10 percent, applies and is not included in its total, so the budget includes it as an estimate ($25,988 on items 1 and 2). The Tibidabo quotation includes an estimated tariff surcharge of $10,725, and the Newport prices include their tariff surcharges.

Room preparation at MSU expense. Building electrical work is not allowable under DURIP. MSU will provide the dedicated circuits for the laser and chiller (Facilities estimate about $8,000) and a door interlock for the Class 4 laser area at its own cost.

Module B fabrication and hydrogen safety. Module B is assembled at MSU from components that are each below the equipment threshold, and the Office of Research Administration will confirm the fabricated-equipment account in writing before the first purchase. The vessel and pump are ordered at award. Hydrogen is not connected until micro-tube choking is verified and the university's hydrogen safety review is approved. The ventilation for the hydrogen cabinet is building work and is paid by MSU; item 15 covers the free-standing cabinet only.

Laser safety. A Class 4 standard operating procedure will be approved by the university's Environmental Health and Safety office before first light, and all users complete laser-safety training. The filtered laser-safe enclosure, the curtains and the eyewear are part of the system (items 10 and 11).

Factory acceptance and training. The PI will travel once to the Light Conversion factory in Vilnius, Lithuania, to witness the factory acceptance test of the laser and OPA before shipment (travel line, $3,500), as the NOFO permits for testing and acceptance. Operator training is delivered on site by the vendor.

Quotations and compliance. Every item is supported either by a recent written vendor quotation or by a documented budgetary estimate whose quotation has been requested. Table B-2 of the Budget Justification lists each quotation with its number, date and validity. Quotations in hand cover the laser and OPA with their extended warranty, the intensifier, the delay stage, the delay generator, the optical tables and the Bragg-grating filters, and parts of items 10, 11, 14, 15 and 16. Their validity runs from 30 to 180 days, so prices will be re-confirmed at award. Light Conversion states that it usually honors quoted prices, and Zaber holds its price through the award. No item is covered telecommunications or video-surveillance equipment under Section 889.

1.11 Estimated useful life and sustainment

The estimated useful life of the system is at least ten years. The optical tables, the spectrograph, the shaper optics, and the Module B vessel, burners and gas-handling hardware have useful lives of 15 years or more; the diode-pumped Yb amplifier and OPA, with field-replaceable pump modules, ten years or more; the camera and intensifier eight to ten years; and the workstation five years. The vendor's standard one-year warranty and the second-year extension requested here (Light Conversion Option 1) cover the laser and OPA for their first two years from shipment. After that, routine maintenance (pump-diode replacement, optics, recalibration) will be paid from the research awards that use the system and from School of Engineering support. The PI and trained graduate students operate the system under written procedures, so the capability does not depend on one person.

1.12 Research-related education

MSU is designated by the State of Maryland as its Preeminent Public Urban Research University and is Maryland's largest Historically Black University. The instrument will strengthen its education of scientists and engineers in disciplines important to DoW in three ways. (1) **Research training.** Graduate students will install, commission and operate the system and use it for thesis research. Undergraduates from the PI's rocketry program will take part in assembly, timing, data acquisition and data reduction through research-credit and senior-design projects, with at least two graduate and four undergraduate students per year. (2) **Course content.** The system supports teaching modules in ultrafast optics, Fourier optics (the pulse shaper is a working Fourier-transform demonstration), spectroscopy, and data analysis on datasets of millions of spectra, for students in mechatronics, physics and electrical engineering courses. (3) **Skills DoW laboratories need.** Students will graduate with hands-on experience in ultrafast lasers, nonlinear spectroscopy and high-rate data handling, the skills used in Navy, Army and Air Force propulsion and diagnostics laboratories, and with publications built on DoW-relevant measurements.

1.13 Ongoing and proposed support for the research

Table 8 lists the ongoing and proposed support for the research that will use the system. The PI's pending ONR Long Range BAA proposal is the research the system most directly serves, and

Module B is the hardware that proposal assigns to this DURIP request. The State of Maryland and U.S. Army rocketry awards support the propulsion program in which the high-speed camera will also be used.

Table 8. *Ongoing and proposed support for research that will use the instrument (PI: Dr. Oluwatobi Busari unless noted).*

Sponsor and number	Title	Amount	Period	Status; use of instrument
Office of Naval Research, Code 352 (copy to Code 351); BAA N0001425SB001	Altitude Compensated Combustion Diagnostics (ACCD) Laboratory; Co-PI Dr. Albert O. Okorogu	$658,586 (research effort; no equipment)	07/2027–06/2030	Pending (submitted 30 Sep 2026); Thrusts 1–4; Module B is its hardware
DoW HBCU/TCU Equipment and Instrumentation; W911NF26S0105	Passive coherence-thermometry instrumentation	$1,292,000	12 months	Submitted 9 Oct 2026; Thrust 4(a); no shared items
U.S. Army	University Outreach in Rocketry	$19,799	09/2024–12/2026	Current; Thrust 4(b)
State of Maryland	Morgan State Rocketry Program	$436,750	08/2023–07/2027	Current; Thrust 4(b)
NASA University Leadership Initiative (Purdue lead)	ASCENT: rotating-detonation combined cycle	$300,000	11/2026–10/2029	Pending; Thrust 1(c)
National Science Foundation (EAGER)	Multipole-resolved chemiluminescence	$300,000	10/2026–09/2029	Pending; Thrust 4(a)

All awards and proposals are listed in the PI's Current and Pending Support. The hardware in this request is not requested from any other sponsor except as disclosed there.

2. Management Approach

This is a single-PI award. Dr. Oluwatobi Busari is responsible for specification, procurement, installation, acceptance, safety, operation and reporting. He commits about 420 hours to installation, integration and acceptance testing, and graduate and undergraduate students contribute about 720 hours, none of it charged to this award, since DURIP does not pay salaries. Dr. Albert O. Okorogu, Associate Professor and Co-PI of the ONR proposal, designs the Module B collection optics (item 18); no salary is requested for him here. Procurement runs through MSU Purchasing and the Office of Research Administration (ORA). The long-lead orders are placed in Month 1 and tracked at a monthly procurement review with ORA until all items are accepted. The university's Environmental Health and Safety office approves the Class 4 laser SOP before first light and the hydrogen safety review before hydrogen is connected, and Facilities completes the room preparation by Month 3 (Section 1.10).

Shared use. Once commissioned, the system is available to other MSU investigators and to DoW collaborators through a shared-use calendar maintained by the PI, with priority for DoW-funded research and a trained-operator requirement for all users. Use, uptime and maintenance are logged and reported in the progress and final reports.

3. Principal Investigator Qualifications

Dr. Oluwatobi Busari (ORCID 0009-0008-6471-3831) is Assistant Professor of Mechatronics Engineering in the Clarence M. Mitchell, Jr. School of Engineering at Morgan State University. He joined MSU in 2021 as Rocketry Research Associate and was appointed to the faculty in 2024. He holds the Ph.D. (2021) and M.S. (2016) in Aeronautics and Astronautics from Purdue University, and the B.S. in Aerospace Engineering and B.A. in Physics (2012) from the State University of New York at Buffalo. His graduate research at Purdue, in the laser-diagnostics group of R. P. Lucht, was carried out in optically accessible, high-pressure combustors. It covered flame stabilization and flow-field structure of reacting jets in vitiated crossflow [13, 14], NO_x reduction through the exit Mach number of an axially staged gas-turbine model combustor [8], the design of an optically accessible secondary combustion zone [15], and natural-gas–hydrogen reacting jets at engine-relevant conditions [16]. He has written the first-principles design basis for the requested system and the density-matrix CARS model used to check it [5]. At MSU he leads the university's liquid-rocket program, funded by the State of Maryland since 2023 and by a U.S. Army university-outreach award, and is developing the Lake Clifton test site. This award draws on his experience in optical-access combustor design, laser-diagnostic modeling and laboratory build-out.

4. Responsibility

Morgan State University is classified R2 (Doctoral Universities: High Research Activity). The university has the resources, and the ability to obtain the resources, to complete this award: the laboratory space described in Section 1.9 exists, the room preparation that DURIP does not fund is committed by the university (Section 1.10), and the PI's current and pending commitments, listed in his Current and Pending Support, leave the uncharged installation effort described in Section 2 available.

The Office of Research Administration administers the institution's sponsored portfolio. In FY2025 it managed 161 new awards and more than $104 million in sponsored-research commitments from NASA, NSF, DoD, DOE/NNSA, NIH, the State of Maryland and other sponsors, including a current U.S. Army award to the PI. MSU maintains active registrations in SAM.gov (UEI KULSKCCZJT27; CAGE 8U547) and Grants.gov, holds a negotiated facilities-and-administrative rate agreement with the Department of Health and Human Services as cognizant agency, and operates the financial, accounting, procurement, property-control and compliance systems required by 2 CFR 200, with an annual Single Audit under 2 CFR 200 Subpart F. Equipment acquired under the award will be tagged and tracked in the university's property-control system. The university is not debarred or suspended. The PI certifies compliance with research-security requirements, including disclosure of all sources of research support and of any participation in a foreign talent-recruitment program, in the attached Senior/Key Person Profile and Current and Pending Support.

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Facilities and Other Resources

Laser laboratory, Room 229, Clarence M. Mitchell, Jr. School of Engineering. The PI's laboratory, where the system will be installed, is a windowless room already fitted with laser barrier curtains. Floor area, available electrical service, cooling water, gas supply and exhaust, and the existing optical and data-acquisition equipment in the room are to be listed here by the PI. The university will add the dedicated circuits for the laser and chiller and a door interlock for the Class 4 laser area (below).

Lake Clifton site. The university-owned former Lake Clifton High School property in Baltimore is being developed by the PI's group for liquid-rocket static-fire testing, under a fire-protection and life-safety review coordinated with the Office of Safety, Health and Environment. It provides exhaust, pump-noise isolation and hydrogen ventilation for injector testing and long-duration hydrogen operation of the Module B vessel. The portable camera and intensifier will also be used there for static-fire imaging.

Machine shop and fabrication. The School of Engineering machine shop provides the milling, turning and wire-EDM capability used for optical mounts, fixtures, the Hencken burner and the injector head.

Computing and data. The dedicated workstation (item 12) acquires and fits spectra. Data are archived on university-managed, backed-up storage.

Institutional support. The Office of Research Administration manages procurement and award administration. The Environmental Health and Safety office approves the Class 4 laser SOP and trains users. Facilities provides the room preparation described in Section 1.10 of the narrative at university expense.

Equipment and work at university expense. Dedicated electrical circuits for the laser and chiller and a door interlock for the laser area (about $8,000), moving the optical tables into Room 229, and the ventilation for the hydrogen cabinet. No government-owned facilities or equipment are used for this research.

Budget (Morgan State University Budget Form)

	MORGAN STATE UNIVERSITY					
UEI: KULSKCCZJT27	**Agency/ Solicitation: Office of Naval Research, FY2027 DURIP, N0001426SF003**					
	Project Title: High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility					
	Project/Budget Period: 06/18/2027-06/17/2028					
MSU Account Code	**A. Senior Personnel**	**Base Salary**	**Year 1**	**Year 2**	**Year 3**	**Total**
02	PI: Dr. Oluwatobi Busari (0.0 SM)	**$ 105,921**	$ -	$ -	$ -	**$ -**
	B. Other Personnel					
02	Graduate Student (GRA)		$ -	$ -	$ -	**$ -**
02	Technician - none		$ -	$ -	$ -	**$ -**
	Personnel Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	C. Fringe Benefits					
02	Fringe benefits @ 9%		$ -	$ -	$ -	**$ -**
01 or 02	Fringe benefits @ 42%		$ -	$ -	$ -	**$ -**
	Fringe Benefits Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	D. Equipment					
11	1. Yb fs amplifier, Light Conversion CARBIDE-40W		$ 220,904	$ -	$ -	**$ 220,904**
11	2. Yb-pumped OPA, Light Conversion ORPHEUS-F		$ 76,164	$ -	$ -	**$ 76,164**
11	3. Dual-channel 4f pulse shaper (LCoS SLM)		$ 35,000	$ -	$ -	**$ 35,000**
11	4. Imaging spectrograph, 320 mm		$ 40,000	$ -	$ -	**$ 40,000**
11	5. High-speed camera, 512 GB (Phantom TMX		$ 264,000	$ -	$ -	**$ 264,000**
11	6. Gated intensifier, Lambert HiCATT25 + gate		$ 152,531	$ -	$ -	**$ 152,531**
11	7. Delay stage, Zaber X-LDQ0300C, 300 mm		$ 8,689	$ -	$ -	**$ 8,689**
11	8. Delay generator, Berkeley Nucleonics 577-8C		$ 7,583	$ -	$ -	**$ 7,583**
11	9. Optical tables, Newport RS2000, with isolators		$ 37,735	$ -	$ -	**$ 37,735**
11	10. Beam optics, power meter, profiler, eyewear		$ 45,000	$ -	$ -	**$ 45,000**
11	11. HEPA/laser-safe enclosure; laser curtains		$ 16,000	$ -	$ -	**$ 16,000**
11	12. GPU acquisition and fitting workstation		$ 12,000	$ -	$ -	**$ 12,000**
11	13. Mod. B: Hencken burner, gas panel (TA-1)		$ 4,690	$ -	$ -	**$ 4,690**
11	14. Mod. B: reduced-pressure vessel (TA-2)		$ 61,300	$ -	$ -	**$ 61,300**
11	15. Mod. B: hydrogen capability (TA-3)		$ 11,500	$ -	$ -	**$ 11,500**
11	16. Mod. B: single-element RDC injector (TA-4)		$ 13,600	$ -	$ -	**$ 13,600**
11	17. Mod. B: spectrometer wavelength ref. (D-1)		$ 14,000	$ -	$ -	**$ 14,000**
11	18. Mod. B: VBG notch filters, OptiGrate (D-2)		$ 16,775	$ -	$ -	**$ 16,775**
	Equipment Sub-total		**$ 1,037,471**	**$ -**	**$ -**	**$ 1,037,471**
	E. Int'l/Domestic Travel					
04	Int'l Travel - PI, acceptance test, Vilnius (1 trip)		$ 3,500	$ -	$ -	**$ 3,500**
	Travel Sub-total		**$ 3,500**	**$ -**	**$ -**	**$ 3,500**
	F. Participant Support Costs					
12	1. Stipends		$ -	$ -	$ -	**$ -**
12	2. Travel		$ -	$ -	$ -	**$ -**
	Participant Support Sub-total		**$ -**	**$ -**	**$ -**	**$ -**
	G. Other Direct Costs					
09	1. Materials and Supplies		$ -	$ -	$ -	**$ -**
09990	2. Publications/Exhibitions		$ -	$ -	$ -	**$ -**
08	3. Consultants		$ -	$ -	$ -	**$ -**
08	4. Computer Services		$ -	$ -	$ -	**$ -**
08191	5. Subaward		$ -	$ -	$ -	**$ -**
12	6. Other - Tuition		$ -	$ -	$ -	**$ -**
08999	7. Other - Event Support		$ -	$ -	$ -	**$ -**
08	8. Other - Extended warranty, laser + OPA		$ 16,320	$ -	$ -	**$ 16,320**
	Other Direct Costs Sub-total		**$ 16,320**	**$ -**	**$ -**	**$ 16,320**
	H. Total Direct Costs		**$ 1,057,291**	**$ -**	**$ -**	**$ 1,057,291**
	Modified Total Direct Costs (MTDC)		**$ 19,820**	**$ -**	**$ -**	**$ 19,820**
15	**I. Indirect Costs (54% MTDC)**		**$ 10,703**	**$ -**	**$ -**	**$ 10,703**
	J. Total Project Costs		**$ 1,067,994**	**$ -**	**$ -**	**$ 1,067,994**

Budget Justification

N0001426SF003

Title: High-Repetition-Rate Femtosecond CARS System with a Sub-Atmospheric Reference-Flame Facility for Time-Resolved Thermometry and Energy-Transfer Kinetics in Naval Propulsion Flows

Budget Period: 06/18/2027 – 06/17/2028

Senior Personnel **TOTAL AMOUNT: $0**

PI: Dr. Oluwatobi Busari **0.0 (SUMR)/yr**

Dr. Busari is an Assistant Professor of Mechatronics Engineering at Morgan State University (MSU) and serves as Principal Investigator. He leads specification, procurement, installation, acceptance testing and safety for the instrument. DURIP does not allow direct salaries for faculty, postdoctoral associates or students, so no salary is requested. His base salary is $105,921 for a 9.5-month appointment; 0.0 summer months are requested, for $0 per year. Dr. Busari contributes about 420 hours to installation, integration and acceptance testing at no cost to the award.

A. Senior Personnel	Year 1	Year 2	Year 3	Total
PI: Dr. Oluwatobi Busari	$0			$0
Senior Personnel Subtotal	**$0**			**$0**

Other Personnel **TOTAL AMOUNT: $0**

No graduate research assistant stipend is budgeted, because student salaries are unallowable under DURIP. Graduate and undergraduate students contribute about 720 hours to installation, integration and acceptance tests, supported from other sources. No program manager or technician is budgeted.

B. Other Personnel	Year 1	Year 2	Year 3	Total
Graduate Student (GRA)	$0			$0
Technician: none	$0			$0
Other Personnel Subtotal	**$0**			**$0**

Fringe Benefits **TOTAL AMOUNT: $0**

MSU fringe charges: For faculty with 9.5-month appointments, fringe charges are budgeted at 42% of the requested salary for the academic year and 9% of the requested salary for the summer. Fringe charges for students and part-time employees are 9% of the requested salary. Because no salary is requested, no fringe benefits are budgeted.

C. Fringe Benefits	Year 1	Year 2	Year 3	Total
PI Busari (SUMR) Fringe @ 9%	$0			$0
Fringe @ 42% (no academic-year salary)	$0			$0
Fringe Benefits Subtotal	**$0**			**$0**

D. Equipment **TOTAL AMOUNT: $1,037,471**

All eighteen items form one instrumentation system for a common research purpose, time-resolved CARS measurement of temperature and molecular state populations, as NOFO Section b.3.i requires. Module A, the 100 kHz fs-CARS diagnostic, is items 1 to 12 ($915,606). Module B, the sub-atmospheric reference-flame facility it measures, is items 13 to 18 ($121,865); it is also the test-article hardware of the PI's pending ONR Long Range BAA proposal, which requests no equipment. All items are purchased at award in Year 1 and are exempt from indirect cost. Items 3, 9, 10 and 11, and all of Module B, are components assembled into the system and are budgeted as equipment; the Office of Research Administration will confirm the fabricated-equipment account. Vendor installation, delivery, and import duty or tariff are part of each item's acquisition cost. Items with a written vendor quotation are budgeted at the quoted price, and the others carry budgetary estimates whose basis is stated below.

1. Yb femtosecond amplifier, Light Conversion CARBIDE-40W-0.4mJ ($220,904). The 1030 nm amplifier (40 W, 400 µJ at 100 kHz, < 250 fs, repetition rate 100 kHz to 1 MHz with an integral pulse picker, chiller and control laptop) is the source of the system. It pumps the OPA and supplies the Stokes beam, and its 100 kHz rate gives one hundred times the measurement rate of 1 kHz CARS instruments. Light Conversion-USA Quotation Q261016-AS1 (19 October 2026, addressed to Dr. Busari, U.S. educational/government discount included): CARBIDE-40W-0.4mJ $190,640; air-freight delivery to the loading dock $3,300; on-site installation and training, including the engineer's airfare and expenses, $7,900. The quotation states that an EU import duty, currently 10 percent, applies and is not included in its total; it is budgeted at 10 percent of the laser price, $19,064. Total $190,640 + $3,300 + $7,900 + $19,064 = $220,904. Delivery is 6 to 8 months after order.

2. Yb-pumped optical parametric amplifier, Light Conversion ORPHEUS-F ($76,164). Converts most of the amplifier output into a tunable 650–900 nm signal (< 55 fs at 800–900 nm; idler 1200–2500 nm) that serves as the broadband CARS pump and probe and reaches the Q-branches of H_2, H_2O, CH_4, N_2, CO and O_2. Same quotation: ORPHEUS-F $69,240, plus the estimated 10 percent import duty, $6,924. Total $76,164.

3. Dual-channel reflective 4f pulse shaper ($35,000). A liquid-crystal-on-silicon spatial light modulator (1272 × 1024) with gratings, optics and fabricated mounts. One channel masks the pump spectrum to select the excited molecules; the other sets a chirped probe of 0.5 to 5 ps. Hamamatsu has identified the X15213-02 (800 ± 50 nm), or its cooled version X15213-02L, as the model for this band and average power, and its quotation has been requested. Budgetary estimate.

4. Imaging spectrograph, 320 mm, aberration-corrected, 1200 l/mm grating ($40,000). Disperses the CARS signal onto the intensified camera (F-mount) with the resolution needed for N_2, H_2 and O_2 Q-branch thermometry. Budgetary estimate based on a public award for a comparable spectrograph (UKRI 2023, £30,453); quotations have been requested from Teledyne Princeton Instruments and Andor.

5. High-speed CMOS camera, 512 GB, Phantom TMX 7510 class ($264,000). Records one CARS spectrum per laser shot at 100 to 200 kHz over a narrow spectral region of interest, and is also used without the laser for high-speed imaging of flames and static fires. Budgetary estimate

from U.S. Army purchase order W911QX24P0128 (two units for $527,855); a quotation has been requested from Vision Research (AMETEK).

6. Gated GaAs image intensifier, Lambert HiCATT25, with gate control ($152,531). Adds gain for single-shot spectra down to about 320 Torr and gates out flame luminosity. Tibidabo Scientific Industries (U.S. distributor for Lambert Instruments) Quotation 126946 V2 (20 October 2026, valid to 20 November 2026): HiCATT25/GaAs/3ns/300kHz/1:1/F-mount $135,430; IC-GC-USB gate control $6,376; estimated tariff surcharge $10,725. Total $152,531. Lead time 22 to 26 weeks, Ex-Works Madison, CT.

7. Motorized delay stage, Zaber X-LDQ0300C-AE53D12-KX15C ($8,689). Sets the probe delay (300 mm of travel covers 2 ns). Zaber Quotation SO73573 (9 October 2026): $8,539 plus shipping $150. The sales tax on the quotation ($521.34) is excluded because MSU is tax-exempt. Price held through the June 2027 award.

8. Digital delay generator, Berkeley Nucleonics 577-8C ($7,583). Synchronizes the laser, the intensifier gate and the camera. Quote 33996 (3 September 2026, valid to 3 December 2026): $7,192.50 plus shipping $390.00 = $7,582.50, budgeted as $7,583.

9. Optical tables, Newport RS2000, with pneumatic isolation ($37,735). One table carries the laser, OPA and shaper; the other the CARS beam overlap, the test article and the detection system. Newport (MKS) Quotation Q-296033 (9 October 2026, valid to 15 November 2026), prices including tariff surcharges: RS2000-510-12 $13,446.65; RS2000-48-12 $8,898.17; two S-2000A-423.5 isolator sets $10,860.32; shipping $4,530. Total $37,735.14, budgeted as $37,735.

10. Beam-delivery optics and optomechanics, power meter, beam profiler, laser-safety eyewear ($45,000). Mirrors, telescopes, lenses, dichroic combiners, filters and mounts that route and overlap the three beams, the power meter and profiler used for alignment and acceptance, and Class 4 laser eyewear. Gentec-EO Quotation 69441-1 (14 October 2026) covers the measurement part, $11,338.70 (50 W thermal detector with USB module, BEAMAGE-4M profiler, 60 W attenuator, stands, service plan, ND set, shipping). The remaining $33,661, for Thorlabs optics and Kentek eyewear, is a budgetary estimate with quotations requested.

11. Laser-table enclosure with HEPA fan-filter units and laser-safe panels; laser-safety curtains ($16,000). Protects the amplifier and OPA optics from dust and contains stray beams. Newport Quotation Q-296033 includes the TE80-510 enclosure at $9,436.86. The HEPA units and laser-safe panels have been specified to Newport with a concept drawing, and the free-standing curtains have been requested from Kentek and Rockwell Laser Industries; the balance of $6,563 is a budgetary estimate.

12. GPU acquisition and spectral-fitting workstation ($12,000). Dedicated to the instrument: streams camera data (up to 512 GB per record) and fits spectral libraries to millions of spectra (24-core processor, 256 GB memory, 32 GB professional GPU, 40 TB storage, 10 GbE). Allowable as computing equipment dedicated to DoW-relevant research. Budgetary estimate; a quotation is being obtained through MSU purchasing.

13. Module B: atmospheric Hencken burner, gas panel, pressure instrumentation (TA-1) ($4,690). A 50 mm micro-tube Hencken burner with mass-flow controllers for CH_4, air, O_2 and N_2, pressure instrumentation and a wavelength reference; the atmospheric anchor flame (1600 to

2400 K) for every measurement. Itemized estimate from the PI; the Alicat mass-flow-controller quotation has been requested, and the burner is fabricated in the MSU machine shop.

14. Module B: reduced-pressure vessel and pressure control (TA-2) ($61,300). A 300 mm inner-diameter × 600 mm 304 stainless vessel with seven port flanges ($18,000); six UV-fused-silica windows ($4,200); a 100 m^3/h chemical-duty dry screw pump ($16,000); a 2 kW exhaust heat exchanger ($2,200); a throttle valve with closed-loop controller ($6,500); capacitance manometers ($4,800); vacuum plumbing ($3,500); burner revisions for reduced pressure ($1,900); and a UV flame detector, interlock PLC and nitrogen purge panel ($4,200). Crystran Quotation APR44409 (16 October 2026) prices the six windows at £7,200 (about $9,635); the item total remains a budgetary estimate until the Kurt J. Lesker (SQ10-0171261) and Busch quotations are received.

15. Module B: hydrogen capability (TA-3) ($11,500). Hydrogen mass-flow controller with CGA 350 two-stage regulator and excess-flow valve ($3,400), two-sensor leak detection with controller ($2,800), a free-standing gas cabinet ($3,500) and a purge panel ($1,800). Applied Energy Systems (Quotation 26957) and PureAire Monitoring Systems (Quotations 1020386 and 1020387) quoted these components on 23 October 2026; the line is held at the estimate until the configuration is selected. The building ventilation for the cabinet is paid by MSU.

16. Module B: single-element rotating-detonation injector (TA-4) ($13,600). A modular injector head in 17-4 stainless steel in three geometries ($8,500), two high-response plenum pressure transducers ($3,200) and a sapphire injector-face window ($1,900). Kulite has given rough-order-of-magnitude prices for its DTL-190 transducer ($1,767 each, five-piece minimum), and its formal quotation has been requested.

17. Module B: spectrometer-based wavelength reference (D-1) ($14,000). A high-resolution fiber spectrometer covering 500 to 1100 nm with Ne, Ar and Hg calibration lamps, the wavelength reference for OPA tuning and for atomic emission lines. It replaces the scanning wavemeter originally considered, which cannot measure broadband femtosecond pulses (Bristol Instruments declined; HighFinesse through TOPTICA confirmed). Budgetary estimate.

18. Module B: volume-Bragg-grating notch filters, OptiGrate BragGrate (D-2) ($16,775). Narrow-band filters at 620, 700 and 780 nm that reject the pump and probe lines and flame chemiluminescence in the collection optics designed by Dr. Albert O. Okorogu. OptiGrate (IPG Photonics) Quotation QD-2026-1014-MGNU-1 (14 October 2026, valid 180 days): three filters at $3,680 each ($11,040), broadband AR coating $5,700, shipping $35. Total $16,775.

D. Equipment	Year 1	Year 2	Year 3	Total
1. Yb femtosecond amplifier, CARBIDE-40W	$220,904			$220,904
2. Yb-pumped OPA, ORPHEUS-F	$76,164			$76,164
3. Dual-channel 4f pulse shaper	$35,000			$35,000
4. Imaging spectrograph, 320 mm	$40,000			$40,000
5. High-speed CMOS camera, 512 GB	$264,000			$264,000
6. Gated intensifier, HiCATT25, with gate control	$152,531			$152,531
7. Delay stage, Zaber X-LDQ0300C	$8,689			$8,689
8. Delay generator, BNC 577-8C	$7,583			$7,583
9. Optical tables, Newport RS2000, with isolation	$37,735			$37,735
10. Beam optics, power meter, profiler, eyewear	$45,000			$45,000
11. HEPA / laser-safe enclosure; curtains	$16,000			$16,000

12. GPU acquisition and fitting workstation	$12,000			$12,000
13. Module B: Hencken burner, gas panel (TA-1)	$4,690			$4,690
14. Module B: reduced-pressure vessel (TA-2)	$61,300			$61,300
15. Module B: hydrogen capability (TA-3)	$11,500			$11,500
16. Module B: single-element RDC injector (TA-4)	$13,600			$13,600
17. Module B: spectrometer wavelength reference (D-1)	$14,000			$14,000
18. Module B: VBG notch filters (D-2)	$16,775			$16,775
Equipment Subtotal	**$1,037,471**			**$1,037,471**

Vendor quotations. The table below lists the quotations behind the equipment and warranty lines (status 4 November 2026). Quotation PDFs are kept with the proposal file.

Item	Vendor and quotation	Date	Valid until	Quoted	Budgeted
1–2	Light Conversion-USA, Q261016-AS1	19 Oct 2026	18 Nov 2026	$271,080 + est. duty $25,988	$297,068
W	Light Conversion-USA, Q261016-AS1, Option 1	19 Oct 2026	18 Nov 2026	$16,320	$16,320
6	Tibidabo Scientific (Lambert), 126946 V2	20 Oct 2026	20 Nov 2026	$152,531 incl. est. tariff	$152,531
7	Zaber Technologies, SO73573	9 Oct 2026	6 Feb 2027; price held to award	$8,689	$8,689
8	Berkeley Nucleonics, 33996	3 Sep 2026	3 Dec 2026	$7,582.50	$7,583
9	Newport (MKS), Q-296033	9 Oct 2026	15 Nov 2026	$37,735.14	$37,735
18	OptiGrate (IPG Photonics), QD-2026-1014-MGNU-1	14 Oct 2026	180 days	$16,775	$16,775
10	Gentec-EO, 69441-1 (power meter, profiler)	14 Oct 2026	30 days	$11,338.70	part of $45,000
11	Newport (MKS), Q-296033 (TE80-510 enclosure)	9 Oct 2026	15 Nov 2026	$9,436.86	part of $16,000
14	Crystran, APR44409 (6 UVFS windows)	16 Oct 2026	30 days	£7,200 (≈ $9,635)	part of $61,300
15	Applied Energy Systems, 26957 (H_2 gas cabinet)	23 Oct 2026		received	part of $11,500
15	PureAire, 1020386 and 1020387 (H_2 leak detection)	23 Oct 2026	120 days	received	part of $11,500
16	Kulite, rough order of magnitude (DTL-190)	14 Oct 2026	n/a	$1,767 each, 5 minimum	part of $13,600
3	Hamamatsu, X15213-02 / -02L	requested	pending		$35,000
4	Teledyne Princeton Instruments; Andor	requested	pending		$40,000
5	Vision Research (AMETEK), Phantom TMX 7510	requested	pending		$264,000
10	Thorlabs optics lot; Kentek eyewear	requested	pending		balance of $45,000
11	Newport HEPA units and panels; Kentek / Rockwell curtains	requested	pending		balance of $16,000
12	Workstation (MSU purchasing)	requested	pending		$12,000
13	Alicat mass-flow controllers; MSU machine shop	requested	pending		$4,690

14	Kurt J. Lesker, SQ10-0171261 (vessel); Busch (pump)	requested	pending		balance of $61,300
17	Spectrometer-based wavelength reference	estimate			$14,000

Travel **TOTAL AMOUNT: $3,500**

One trip in Year 1 for the PI to Light Conversion in Vilnius, Lithuania, to witness the factory acceptance test of the laser and OPA before shipment, as the NOFO allows for testing and acceptance. Operator training is delivered on site at MSU and is included in the laser quotation. Six days and five nights; airfare is a U.S.-flag or code-share economy fare, in compliance with the Fly America Act.

Purpose	Airfare	Lodging	Taxi / Parking	Meals	Registration	Total
Factory acceptance test, Vilnius	$1,400	$240 x 5 = $1,200	$163	$134 x 4 + $100.50 x 2 = $737	$0	**$3,500**

Lodging and Meals & Incidentals follow the U.S. Department of State per-diem rates for the host city.

E. Int'l/Domestic Travel	Year 1	Year 2	Year 3	Total
International Travel — PI (factory acceptance test, Vilnius)	$3,500			$3,500
Travel Subtotal	**$3,500**			**$3,500**

Participant / Trainee Support Costs **TOTAL AMOUNT: $0**

Participant support costs are not budgeted.

F. Participant Support Costs	Year 1	Year 2	Year 3	Total
1. Stipends	—	—	—	—
2. Travel	—	—	—	—
Participant Support Subtotal	**$0**	**$0**	**$0**	**$0**

Other Direct Costs **TOTAL AMOUNT: $16,320**

Other — Extended warranty, laser and OPA: a one-year extension of the vendor's standard warranty (the second year after shipment), covering parts, labor and on-site service for the amplifier and OPA. Light Conversion Quotation Q261016-AS1, Option 1, $16,320, purchased before the first-year warranty expires. A third-year extension (Option 2, $23,940) is not requested. DURIP allows operation and maintenance costs, including extended warranties. No materials, publication, consultant, computer-service, subaward, tuition or event costs are requested.

G. Other Direct Costs	Year 1	Year 2	Year 3	Total
1. Other — Extended warranty, laser and OPA (2nd year)	$16,320			**$16,320**

Other Direct Costs Subtotal	**$16,320**			**$16,320**

Indirect Costs **TOTAL AMOUNT: $10,703**

Morgan State University's federally negotiated indirect cost rate for on-campus research is 54% of modified total direct costs (MTDC). MTDC excludes student tuition, equipment, and the portion of any sub-award in excess of $25,000. As DURIP requires, all equipment is exempt from indirect cost, so the base is the travel and the extended warranty: 0.54 × $19,820 = $10,702.80, rounded to $10,703.

Total Project Costs **TOTAL AMOUNT: $1,067,994**

	Year 1	Year 2	Year 3	Total
H. Total Direct Costs	**$1,057,291**			**$1,057,291**
Modified Total Direct Costs (MTDC)	**$19,820**			**$19,820**
I. Indirect Costs (54% MTDC)	**$10,703**			**$10,703**
J. Total Project Costs	**$1,067,994**			**$1,067,994**